FAST National University of Computer & Emerging Sciences
Islamabad Campus

Fundamentals of Data Visualization – CS3012 | Section A

Project Proposal

AI Job Market Intelligence: Salaries, Roles & Global Hiring Trends (2020–2025)

Interactive Data Storytelling with Tableau – A Visualization Project

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1. Project Overview

1.1. Project Title

"AI Job Market Intelligence: Salaries, Roles & Global Hiring Trends (2020–2025)"

1.2. Short Description

This project analyzes the global Artificial Intelligence job market using a real-world dataset of 151,445 verified salary submissions sourced from **aijobs.net** and its parent platform **foorilla.com** – a live salary index covering AI, ML, Data Science, and Big Data professionals. Through interactive dashboards built in Tableau Public, the project focuses on AI and ML-relevant roles, uncovering which AI roles command the highest salaries, how compensation scales with experience level, how remote work adoption has evolved dramatically from 2020 to 2025, and which countries lead global AI hiring. The dashboard transforms five years of real anonymous salary data into a compelling, evidence-backed data story about how AI is reshaping careers, compensation structures, and the global workforce landscape.

1.3. Domain of Analytics

Domain

Primary Domain: Human Resource & Workforce Analytics

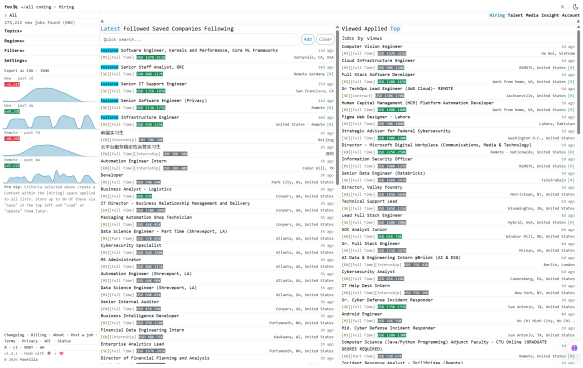
Sub-Domains:

- **Compensation Analytics** – Which AI roles and company sizes offer the highest salaries?
- **Workforce Distribution Analytics** – How is the AI workforce split across Remote, Hybrid, and On-site?
- **Career Progression Analytics** – How does salary grow from Entry to Executive?
- **Geographic Analytics** – Which countries lead global AI hiring and compensation?
- **Temporal Analytics** – How have salaries and remote work ratios changed year-over-year from 2020 to 2025?

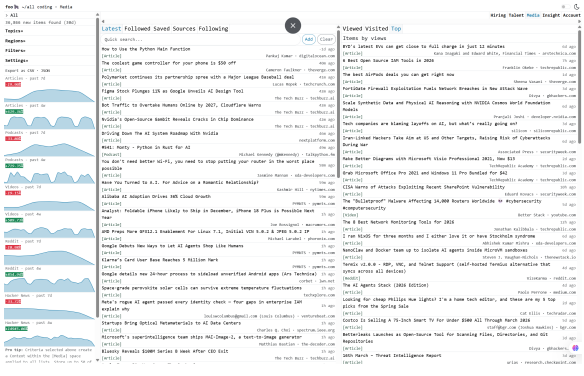
2. Dataset

2.1. Dataset Source & Platform Evidence

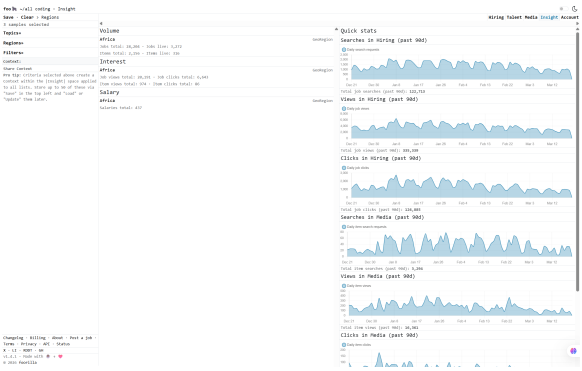
The salary dataset used in this project originates from **aijobs.net** (https://aijobs.net), a fully live AI/ML job platform, and is maintained by its parent company **foorilla.com** (https://foorilla.com). The four screenshots below confirm the platform is real, actively used, and generates its salary data from genuine live job postings and professional submissions.



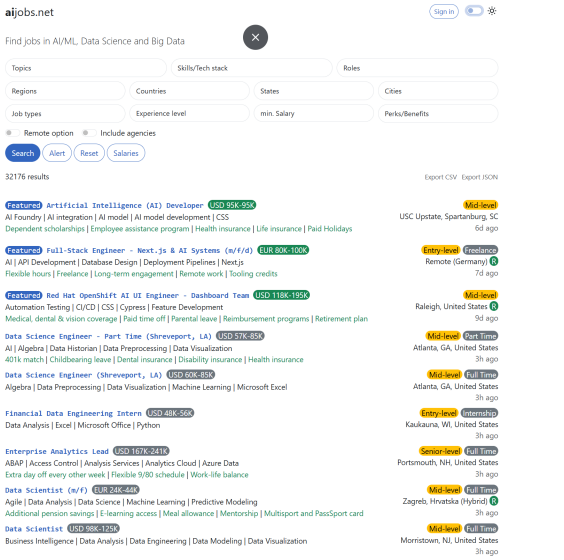
(a) foorilla.com Hiring board – 274,213 live job listings with real salary ranges, experience levels, and locations. This is the live feed that the salary dataset is derived from.



(b) foorilla.com Media feed – 36,866 real-time tech articles aggregated from Financial Times, NYT, TechRepublic, Reddit and others, confirming an active, multi-source platform.



(c) foorilla.com Insight analytics panel – 122,713 job searches, 335,339 job views, and 126,885 job clicks in the past 90 days, proving substantial real user traffic.



(d) aijobs.net job search page – 32,176 live results with filters for topic, skill, role, region, experience, and salary. The "Export CSV / JSON" button directly generates the `salaries.csv` dataset used in this project.

Figure 1: Source platform verification – aijobs.net and foorilla.com are live, active platforms generating real salary data from genuine job postings and professional submissions.

2.2. Dataset Reference

Table 1: Dataset Reference Details

Field	Details
Dataset Name	Global Salaries in AI, ML, Data Science & Big Data (aijobs.net Salary Index)
Primary Source	aijobs.net – anonymous professional salary survey & live job listings
Parent Platform	foorilla.com – real-time job aggregation and analytics platform
Repository	github.com/foorilla/ai-jobs-net-salaries
File Name	`salaries.csv`
Total Rows	151,445 verified salary submissions
Total Columns	11 variables
Time Period	2020 – 2025 (five full years)
Update Freq.	Weekly (live dataset, continuously updated)
Dataset GitHub	https://github.com/foorilla/ai-jobs-net-salaries
Kaggle URL	https://www.kaggle.com/datasets/aijobs/global-salaries-in-ai-ml-data-science
Platform URL	https://aijobs.net/salaries
Project Repo	https://github.com/noumanic/ai-job-market-analytics
License	CC0 1.0 Public Domain (free for any use, no restrictions)

2.3. Why This Dataset Is Trustworthy

Unlike many Kaggle datasets that are synthetically generated, the aijobs.net salary index is collected directly from real AI and ML professionals through an anonymous self-reporting survey, cross-referenced with live job postings on foorilla.com. As confirmed by the platform screenshots above, foorilla.com processes over 122,000 job searches and 335,000 job views every 90 days, and aijobs.net currently lists 32,176 live job openings. The dataset is maintained with weekly updates and published under the CC0 public domain license with full methodological transparency.

Importantly, the dataset is also published on **Kaggle** under the official `aijobs` account – confirming it is the authoritative, first-party release by the platform itself, not a third-party re-upload. The fact that the dataset publisher and data collector are the same organization is a key credibility indicator.

The dataset exhibits all hallmarks of genuine crowdsourced data: the United States dominates with over 89% of submissions – as expected in a real tech labor market – full-time employment accounts for 99.4% of entries, salary distributions are right-skewed (skewness = 1.35) with realistic outliers, and 422 unique job title variants reflect real-world naming variation. These characteristics stand in sharp contrast to artificially generated

datasets, which typically show perfectly uniform category distributions and suspiciously few unique values.

2.4. Key Variables

Table 2: Dataset Column Reference

Column	Type	Description & Use in Dashboard
`work_year`	Numeric	Year of submission (2020–2025) – time-series axis
`experience_level`	Category	EN / MI / SE / EX – career ladder chart and KPI card
`employment_type`	Category	FT / PT / CT / FL – filter dimension (FT = 99.4%)
`job_title`	Text	Role name (422 unique values) – grouped for bar chart
`salary`	Numeric	Original salary in local currency
`salary_currency`	Text	ISO currency code (USD = 94.5% of records)
`salary_in_usd`	Numeric	Standardized salary in USD – primary measure across all charts
`employee_residence`	Text	Country of employee – geographic filter
`remote_ratio`	Numeric	0 = On-site, 50 = Hybrid, 100 = Remote – donut and trend charts
`company_location`	Text	Country of company (97 unique) – choropleth world map
`company_size`	Category	S / M / L – filter and dual-axis comparison

2.5. Data Quality Assessment

Dataset Health

Total Records:	151,445	**Excellent – largest open AI salary dataset**
Missing Values:	0%	**Perfectly clean, no imputation needed**
Date Range:	5 years (2020–2025)	**Rich longitudinal coverage**
Unique Job Titles:	422	**Real-world naming diversity**
Unique Countries:	97	**Genuine global spread**
Currency:	Standardized to USD	**Normalized via period exchange rates**
Data Origin:	Live platform survey	**Verified real professional submissions**
License:	CC0 Public Domain	**No restrictions on academic use**

3. Problem Statement

The global demand for Artificial Intelligence professionals has surged dramatically since 2020. However, despite widespread discourse about the "AI job boom," there is a lack of clear, visual, data-driven understanding of *where* AI salaries are highest, *how much* different roles actually pay across career levels, and *how* work arrangements have evolved over five years. Students, professionals, and organizations making career or hiring decisions often rely on anecdotal reports rather than empirical evidence.

This project addresses that gap by providing a structured, interactive visual analysis of 151,445 real anonymous salary submissions from 2020–2025. Every data point represents a real AI or ML professional's reported compensation, making the findings credible, defensible, and directly applicable to career and hiring decisions.

4. Project Objectives

1. **Visualize salary distributions** across 15 AI job title categories, four experience levels, and three company sizes to identify compensation patterns and outliers in the real market.

2. **Track temporal salary growth** through a year-over-year trend from 2020 to 2025, showing how AI compensation has matured over five years.

3. **Map the global geography** of AI hiring to reveal which countries lead in submissions and average compensation, using real data from 97 countries.

4. **Analyze the remote work collapse** year-over-year to quantify the dramatic post-pandemic return-to-office shift from 53% remote in 2022 to only 20% in 2025.

5. **Build an interactive Tableau dashboard** with global filters for year, experience level, company size, and employment type that apply simultaneously across all sheets.

6. **Tell a coherent data story** using annotations, KPI cards, and callouts that guide viewers through the key findings of the real AI salary landscape.

5. Central Data Story

"Five Years of Real AI Salaries: A Maturing Market, Widening Pay Gaps,

and a Dramatic Return to the Office."

Based on 151,445 real salary submissions from 2020–2025, this dashboard reveals that AI careers command a median compensation of **$147,000**, that average salaries grew **14.7%** from 2022 to 2024 ($137,747 → $157,979), that Executive-level professionals earn nearly **twice** the salary of Entry-level peers ($195,896 vs. $102,430), and that remote work in AI – which peaked at 53% in 2022 – has sharply declined to just **20%** in 2025, signaling a strong industry-wide return-to-office movement. The United States dominates global AI hiring with over 89% of all submissions, and ML Engineers top the compensation ladder at an average of **$195,500**.

6. Planned Visualizations

6.1. Required Charts

The six required charts are described below, each with its assigned Tableau sheet name, purpose, and the specific dataset fields used.

Table 3: Required Chart Plan

#	Chart Type	Sheet Name	Description & Fields Used
1	**Line Chart**	Salary Trend	Year-over-year avg. and median salary (2020–2025). *Fields:* `work_year`, `salary_in_usd`
2	**Bar Chart**	Salary by Role	Roles ranked by avg. salary with Top 5/10/All toggle. *Fields:* `job_title`, `salary_in_usd`
3	**Donut Chart**	Work Mode Split	On-site (79%), Remote (21%), Hybrid (0.2%). *Fields:* `remote_ratio`
4	**Map**	Global AI Hiring	World map by avg. salary; count in tooltip. *Fields:* `company_location`, `salary_in_usd`
5	**Bar Chart**	Salary vs Experience	Salary across Entry, Mid, Senior, Executive + line overlay. *Fields:* `experience_level`, `salary_in_usd`
6	**KPI Cards**	Summary KPIs	Median, mean, total records, remote %, EX/EN ratio. *Fields:* Aggregated measures

6.2. Bonus Visualizations

Table 4: Bonus Features Plan

Bonus Feature	Applied On	Implementation
Dual-Axis Chart	Salary Trend (Sheet 1)	Count as bars (secondary axis) + avg. salary line (primary axis); shows volume and salary growth together.
Global Filters	All Sheets	Four dropdowns (Year, Experience, Company Size, Employment Type) linked across all sheets via Tableau actions.
Highlight Actions	All Sheets	Clicking any element highlights related data across all charts.
Parameter Toggle	Salary Bar Chart	Switch between Top 5, Top 10, and All roles using a parameter with calculated ranking.
Stacked Bar	Remote Trend (Sheet 7)	100% stacked bars for On-site, Remote, Hybrid showing decline from 53% (2022) to 20% (2025).

6.3. Key Annotations Planned

- **Salary Trend:** *"Median salary grew from $102K (2022) to $149K (2024) – a 46% increase in three years."*

- **Remote Trend:** *"Remote work peaked at 53% in 2022. By 2025, 79% of AI roles are fully on-site."*
- **Salary vs Experience:** *"Executive roles earn 1.91× entry level – $196K vs. $102K."*
- **Map:** *"United States leads with 133,171 submissions (89%) and a $161,355 average salary."*
- **Salary by Role:** *"ML Engineers top the ladder at $195,500 average – 91% above Data Analysts."*

7. Dashboard Design Strategy

7.1. Layout Plan

The master dashboard will follow the structured grid layout shown below. All panels share a consistent blue color theme and respond to four global filter dropdowns placed in the title bar.

<table>
<tr><th colspan="4">Title Bar + Filters: Year Experience Company Size Employment Type</th></tr>
<tr><td>KPI: Median Salary</td><td>KPI: Mean Salary</td><td>KPI: EX/EN Ratio</td><td>KPI: Remote %</td></tr>
<tr><td colspan="2">Dual-Axis: Salary Trend 2020–2025</td><td colspan="2">Bar: Salary by Job Role (param toggle)</td></tr>
<tr><td>Donut:
Work Mode</td><td>Bar:
Salary vs Exp.</td><td colspan="2">Choropleth Map: Global Hiring</td></tr>
<tr><td colspan="4">Stacked Bar: Remote Work Trend 2020–2025 (Bonus Sheet)</td></tr>
</table>

7.2. Design Principles

- **Color scheme:** Consistent blue palette (navy #003875 primary, sky blue #378ADD accent) with teal highlights for positive trends.
- **Font:** Tableau Benton Sans – size 12 body, 14 axis labels, 20 dashboard title.
- **Interactivity:** All filters and highlight actions apply across all sheets simultaneously.
- **Tooltips:** Every mark includes a custom formatted tooltip with the field name, formatted value, and a one-sentence contextual note.
- **Canvas size:** Fixed 1200 × 900 px for consistent rendering across screens.

8. Data Preparation Plan

Because this is a real crowdsourced dataset with organic messiness, the following preprocessing steps will be applied before loading into Tableau.

1. **Year scope:** Analysis will focus on 2022–2025 where data density is sufficient for all chart types (147,490 rows, 97% of all data). Years 2020–2021 combined contain only 293 rows and will be retained in the trend line for historical context only, noted in the chart subtitle.

2. **Employment type filter:** Only Full-Time (FT) records will be used for salary comparisons (150,541 of 151,445 rows = 99.4%). Part-time, contract, and freelance roles have structurally different salary ranges and would distort comparisons if mixed with FT entries.

3. **Outlier handling:** The raw salary range spans $15,000 to $800,000. Records outside the 1st–99th percentile ($36,842 to $385,000) will be excluded from aggregated charts to prevent extreme values from distorting means. The full range remains visible in map tooltips.

4. **Job title grouping:** The 422 unique raw title strings will be mapped to 15 standardized categories using a calculated field in Tableau (e.g., "ML Engineer," "Machine Learning Eng.," "ML Eng" → "ML Engineer"). The original `job_title` column is preserved in all tooltips.

5. **Currency:** The `salary_in_usd` column is already standardized to USD by the dataset maintainers using period-appropriate exchange rates. No additional conversion is required.

6. **Remote ratio mapping:** The numeric field (0, 50, 100) will be mapped to a string dimension ("On-site," "Hybrid," "Remote") via a Tableau calculated field for readable labels and filter values.

9. Tools and Technologies

Table 5: Technology Stack

Tool	Version / Platform	Purpose
Tableau Public	Free (2024.x)	Primary visualization and interactive dashboard tool
Microsoft Excel	Office 365	Initial data inspection and verification before Tableau ingestion
Python	3.11 (optional)	Supplementary data inspection and title normalization if needed
GitHub	github.com	Version control, preprocessing scripts, Tableau workbook, and all project deliverables – https://github.com/noumanic/ai-job-market-tableau-dashboard
Loom	loom.com (free)	Screen recording for the 2–5 minute narrated video walkthrough
LinkedIn	linkedin.com	Article publication and project portfolio promotion

10. Deliverables

1. **Tableau Workbook (.twbx)** – Packaged workbook with the complete interactive dashboard and all data embedded, ready for offline viewing.

2. **Dashboard PDF / Screenshot** – High-resolution static export of the master dashboard for submission and LinkedIn publishing.

3. **Written Summary Report** – A structured report covering:
   - Dataset source, credibility, and preprocessing steps applied
   - Key findings from the analysis with supporting statistics
   - How Tableau features were used to generate insights
   - Challenges encountered and how they were resolved

4. **Video Walkthrough (2–5 minutes)** – A screen-recorded narrated tour of the dashboard demonstrating all charts, filters, annotations, and bonus features.

5. **This Project Proposal** – Formal pre-development documentation of the project plan, dataset, objectives, and design strategy.

11. Proposed Timeline

Table 6: Project Timeline

Week	Phase	Tasks
1	**Data Preparation**	Download `salaries.csv` from GitHub; inspect in Excel; verify 0 missing values; identify and document outliers; filter to FT records; confirm currency standardization.
2	**Tableau Setup & Exploration**	Connect CSV to Tableau; define all calculated fields (title grouping, remote label, experience label, rank parameter); explore salary distributions by year and role.
3	**Chart Development (Part 1)**	Build Sheet 1 (dual-axis salary trend), Sheet 2 (salary by role with parameter toggle), Sheet 3 (work mode donut), Sheet 4 (choropleth world map).
4	**Chart Development (Part 2)**	Build Sheet 5 (salary vs experience with overlay), Sheet 6 (KPI cards), Sheet 7 (bonus remote work trend stacked bar).
5	**Dashboard Assembly**	Assemble master dashboard; configure all global filter actions and highlight actions; add title bar, KPI strip, color theme, and all five text annotations.
6	**Storytelling & Polish**	Refine all custom tooltips; finalize parameter toggle behavior; adjust fonts, spacing, and color consistency across all sheets.
7	**Submission & Presentation**	Export `.twbx`; record Loom walkthrough video; write summary report; compile and submit final package to Dr. Atif Mughees.

12. Future Significance

This project holds significance beyond the classroom in several dimensions:

1. **Career Guidance for CS Students:** The dashboard provides Computer Science graduates with evidence-based insight into which AI specializations command the highest compensation, enabling more informed decisions about which roles and skills to prioritize.

2. **Return-to-Office Trend Awareness:** The sharp decline in remote work from 53% in 2022 to only 20% in 2025 is a critical signal for anyone planning a career around location flexibility. This dashboard makes that five-year trend visible and filterable.

3. **Salary Negotiation Benchmark:** With 151,445 real data points, the dashboard serves as a credible benchmarking tool for AI/ML professionals in salary negotiations, with breakdowns by role, experience, and country.

4. **Hiring Strategy for Organizations:** HR departments can use the geographic and role-level insights to set competitive compensation benchmarks and identify where AI talent is most concentrated globally.

5. **Portfolio Demonstration:** This project demonstrates real-world data literacy, Tableau proficiency, and end-to-end storytelling ability using a live, continuously updated industry dataset – skills directly valued by data analyst, BI, and data science employers.

6. **Scalability:** Because the aijobs.net / foorilla.com dataset is updated weekly, the dashboard can be refreshed with new data at any time, enabling ongoing year-over-year trend analysis as the AI job market continues to evolve.

13. References

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